

Table of Contents

Acronyms List:	1
Abstract	2
Introduction	3
1.1 Background	3
1.2 Overview	4
1.3 Objectives	5
Literature Review:	7
3. Project Description	16
3.1 System Requirements	16
3.2 Libraries Used	17
3.3 Techniques Used.....	18
4. Methodology	20
4.1 Planned Methodology & Data Sources	20
4.1.1 Planned Steps to be Conducted.....	20
4.2 Data Sources	22
4.2.1 Asian Based Traffic Signs Dataset.....	22
4.2.2 German Traffic Sign Recognition Benchmark (GTSRB).....	24
4.2.3 Indian Traffic Signs Dataset.....	25
4.3 Class Distributions.....	27
4.4 Actual Implementation	30
4.4.1 Data Preprocessing and Image Resizing.....	30
4.4.2 Model Architecture	31
4.4.3 Data Augmentation	31
4.4.4 Training and Hyperparameter Tuning	32
4.4.5 Model Evaluation	32
4.4.6 Real-Time Testing.....	33
4.5 Logic Diagram of the Final Model.....	33
4.5.1 Logic Diagram	33
4.5.2 Explanation of the Logic Diagram	34
5. Results	35

5.1 Overview	35
5.2 Performance on the Asian Dataset	35
5.3 Performance on the German Dataset.....	36
5.4 Performance on the Indian Dataset (DS3).....	37
5.5 Mixed Dataset Testing.....	38
5.6 Final Model Selection and Refinement (German Dataset - DS2)	38
5.7 Real-Time Prediction Testing	40
6. Discussion.....	41
6.1 General Overview of Results	41
6.2 Comparative Performance Across Datasets	42
6.3 Interpretation of Results by Traffic Sign Group	43
6.4 Limitations	45
7. Future Work	46
8. Conclusion	49
References:	52
Literature review papers:.....	52
Additional Reference papers & conference:.....	53

TABLE OF FIGURES:

FIGURE 1:LR DATA SET IMAGES	7
FIGURE 2:LR ACTUAL AND PROCESSED IMAGES	8
FIGURE 3:LR LOGIC DIAGRAM OF THE MODEL	11
FIGURE 4:LR PLANNING PROCESS OF THE PROJECT	12
FIGURE 5:LR LOGIC DIAGRAM	15
FIGURE 6: DS1 TRAFFIC SIGNAL AHEAD WARNING SIGN	23
FIGURE 7:DS1 SPEED LIMIT 30	23
FIGURE 8: PRIORITY ROAD AHEAD.....	25
FIGURE 9:DS2 SPEED LIMIT 30.	25
FIGURE 10: CATTLES AHEAD	26
FIGURE 11:DS3 SPEED LIMIT 30	27
FIGURE 12:DS1 IMAGES DISTRIBUTION BY CLASS	28
FIGURE 13:DS2 IMAGES DISTRIBUTION BY CLASS	29
FIGURE 14:DS3 IMAGES DISTRIBUTION BY CLASS	30
FIGURE 15: ALL PLANNED AUGMENTATION, IMAGE FROM DS3	31
FIGURE 16: PROJECT LOGIC DIAGRAM	33
FIGURE 17: MODEL 1 PROGRESS	36
FIGURE 18: MODEL 2 PROGRESS	37
FIGURE 19: MODEL 3 PROGRESS	37
FIGURE 20: FINAL MODEL PROGRESS	39
FIGURE 21: INCONSISTENT PREDICTION FROM VIDEO FRAME	41
FIGURE 22: COMPARISON OF ACCURACY BY GROUP.	43
FIGURE 23: COMPARISON OF NO. OF IMAGES PER GROUP.	44

Acronyms List:

CNN - Convolutional Neural Network

GTSRB - German Traffic Sign Recognition Benchmark

YOLO - You Only Look Once

ROI - Region of Interest

RNN - Recurrent Neural Network

AR - Augmented Reality

UI - User Interface

DS - Dataset

EDA - Exploratory Data Analysis

AI - Artificial Intelligence

SGD - Stochastic Gradient Descent

Adam - Adaptive Moment Estimation (Optimizer)

GPU - Graphics Processing Unit

TSR – Traffic Sign Recognition

AV - Autonomous Vehicle

Abstract

Traffic sign recognition is one of the most essential tasks that self-driving cars need to perform optimally while on the road. The present work proposes an ML methodology for traffic sign identification and classification from multiple datasets using CNNs. In implementing this project, training models incorporated datasets of traffic signs from different regions which includes Germany and India with primary consideration to the variability across different language and culture.

There are different CNN structures that have been proposed, including attempts to solve problems such as overfitting of data, compatibility of datasets, and computational complexity. Data augmentation was used to enrich model's Data set while hyperparameter's search was used to attain the best performing model's settings. The final model was also tested not only from static images but also in dynamic environment such as from real-time video feed.

The model trained on German Traffic Sign Recognition Benchmark proved to be the superior model during training of all the models with a validation accuracy of over 99%. However, certain limitations were still apparent with the approach especially when the images are not cropped to focus on the region of interest and there may be occasional misclassification. Even with these difficulties, the results show that CNNs can be used for traffic sign recognition and can laid the basic for future work such as using enhanced models like YOLOv6 for real-time detection.

This work will help the future development of the AV and cloud control or driving optimization technologies improve the technologies through the demonstration of adaptation of TSR models to different geographical locations as well as assessing the efficacy of the Automotive industry in conditions that approximate real-world scenarios. Future work will be oriented to improving real-time detection and embedding the models to edge computing platforms.

Introduction

1.1 Background

Autonomous Vehicles:

Self-driving cars otherwise known as autonomous vehicles are part of the modern technology that has been incorporated in transport systems today. AI and machine learning coupled with computers enable these vehicles to both detect and drive autonomously without any help from human beings. There are several reasons as to why there has been increased effort towards the development of self-driving cars: Improved transportation systems – human error is one of the leading causes of road accidents. This is an exciting technology that has the capability of changing the automotive industry and therefore the future of roads and transport, with hope for safer roads, efficient traffic congestion and transport for all.

The latter is true because autonomous cars need to be able to sense and understand the environment they are operating in. This include identifying other car, people, the road surface and above all the signs. A traffic sign offers information that assists vehicles to make correct decisions when on the road regarding whether to stop, give way or even maintain or reduce the set speed limit. In the case of self-driving cars, if traffic signs are not well understood, the cars can be a source of danger to users and other road users, The project also tells the importance of Traffic Sign Recognition systems.

Traffic Sign Recognition:

Traffic sign recognition is one of the crucial perception modules of an autonomous vehicle. It enables the vehicle to understand the legal requirements of traffic management so as to maintain legal and safe standards. Information signs include the maximum allowable speed, signs that warn of changes in the traffic conditions, and signs which instruct the driver to stop or give way. Lack of correct perception of these signs may expose one to risky conditions such as traffic infringement, or accidents. Hence, there is the necessity to ensure that the traffic sign recognition for self-driving cars is precise.

Nevertheless, there are some difficulties when it comes to traffic sign recognition. The traffic signs are different based on various countries and areas, and due to appearance aspect, there may be a difference in colour, shape, or size because of variation in lighting conditions, weather conditions and occlusions of the sign. Also, traffic signs are usually inscribed in several languages, which adds another level of complication and difficulty faced by autonomous systems that need to read and understand them properly. Therefore, traffic sign recognition systems require much flexibility and reliability to perform optimally in different contexts or region.

CNNs and Machine Learning:

CNN has recently become extremely popular for tasks involving visual recognition, so using it for traffic sign recognition is reasonable. CNNs are a type of deep learning technology with the application of processing and analysing visual data, especially pictures. Due to the application of several layers of convolutional filters, the CNNs can extract and learn overly complex features of image, whereby the CNNs are very capable of recognizing patterns as well as objects. Which makes them suitable for jobs such as traffic sign detection in which ability to differentiate between several types of signs is desirable.

The strength of CNNs is that it is formulating features from raw image data upon its own without any explicit form of feature extraction. This makes the CNNs able to work with large and varied datasets and therefore make traffic sign recognition to be efficient. To achieve this, CNN models are trained on datasets from different regions hence developing this traffic sign recognition system that is versatile of the environment and language used.

1.2 Overview

This thesis explores the development of a traffic sign recognition system using Convolutional Neural Networks (CNNs) to address the needs of autonomous vehicles. The goal of the project is to create a robust model that can accurately classify traffic signs across different regions, including Germany, India, and Asia. The use of diverse datasets

ensures that the models developed are adaptable to various road environments and traffic sign standards, which is crucial for the global deployment of autonomous vehicles.

To achieve this, the project incorporates the German Traffic Sign Recognition Benchmark (GTSRB), the Indian Traffic Signs Dataset, and the Asian Traffic Signs Dataset. These datasets provide traffic signs of different classes/visibility/shape and different languages. Overall, the study will compare the CNN models across the multilingual aim to determine the success and versatility of CNN models in multilingual languages by evaluating several datasets for sign types and road conditions.

The essence of this work is based on the enhancement and optimization of CNN architectures. Several models were produced, and each of them is suitable for the characteristics of the traffic sign datasets. The overfitting problem that characterizes deep learning was countered in the proceeding with the use of data augmentation techniques in training the models. The hyperparameters were also tuned to makes the model generalize the real data environment and reduce the overfitting of the models.

Besides using static images for training the models, the project also experimented with the models in real time video inputs to assess their usability in real time scenarios. This is important in evaluating the usability of the models since self-driving cars are expected to make real-time decisions depending on the traffic signs they come across on the road.

Consequently, the findings of the present research prove the effectiveness of CNNs in traffic sign recognition and outline the issues inherent to transferring models across geographical locations. As mentioned earlier, German Traffic Sign model demonstrated the highest level of accuracy, but for the refinement of real-time recognition particularly in video feed, some improvements are recommended.

1.3 Objectives

This research aims at developing models that are sensitive to variations in the specific traffic sign regulations and language in distinct regions. The specific objectives of the project are as follows:

1. **Develop CNN-Based Models:** Develop and implement CNN (Convolution Neural Network) models on dataset of different area like Germany, India and Asia. The

idea is to create models that can learn the features specific to traffic signs that can be extracted from those regions.

2. **Address Overfitting through Data Augmentation:** along with initial Data preprocessing has to be performed, including application of various augmentation like zoom, grayscale, brightness shift and contrast shift should be done in order to avoid overtraining of the models under consideration. This will help the models to better generalize over unseen data and would help the model in real-world scenario.
3. **Optimize Model Performance through Hyperparameter Tuning:** Perform hyperparameter tuning to refine the models' performance by using GridSearch library. Optimize the models with the help of hyperparameter tuning. This involves such processes like changing of learning rates, altering batch size, and changing of network architectures to increase the accuracy on the cost of the loss.
4. **Evaluate Models in Real-Time Environments:** For the final evaluation of the performance of the model, feed a real-time video in order to see how well the model perform in a real-life setting. This step is input into the consequent recognition procedure to evaluate how traffic signs can be recognized by the models in real life use.
5. **Explore Transfer Learning:** Utilize pre-trained models like MobileNet to increase TSR accuracy especially when working with difficult datasets and ascertain the advantages of transfer learning in traffic sign recognition.

Literature Review:

Below there's literature review of 10 paper which could give a bit more insight and understanding of similar types of projects and their approach

Paper 1 - "Traffic Sign Recognition with Multi-Scale Convolutional Networks"

This paper describes the application of multi-scale Convolutional Neural Networks (CNNs) in recognition of traffic signs. The authors propose a new method based on CNNs which learns the features from the traffic sign images without the necessity to extract the features from the images manually. Multi-scale CNNs are especially useful in visual recognition because they are able to take inputs in different resolutions making it possible for the model to determine local and global features of images. This is especially important for traffic sign recognition because in the real world traffic signs of several types may be at various distances from the car that performs the recognition and thus may be at diverse sizes and rotated in several ways.



Figure 1:LR Data set images

The work also shows that CNNs are suitable for traffic sign classification based on the results of the assessment provided by the German Traffic Sign Recognition Benchmark (GTSRB). This model has been tuned on thousands of images and demonstrates good generalization, and therefore, could be applied in practice. This work is one of the first to show that deep learning, particularly CNNs, can be used for traffic sign recognition and notes that this leads to the possibility of replacing methods that have relied on hand-designing features. the contribution of this paper has paved for future improvements utilizing deep learning techniques for self-driven cars.

Paper 2 - "Real-Time Traffic Sign Recognition from Video by Class-Specific Discriminative Features"

In the contribution of this paper, the researchers put forward a method dealing with the traffic sign recognition in real time based on class-specific discriminative features. Their method is based on identifying traffic signs from video stream, which is one of the important real-life applications especially if it is applied in the context of autonomous driving where speed is an important factor. The authors then present a method that focuses on discriminative features that belong to certain traffic sign classes, whereby the model is then able to classify several types of signs well.



Figure 2:LR actual and processed images

Another important contribution of this work is the real-time processing constraint, which has been incorporated into the feature extraction and classification blocks so as to allow for the integration of the system in real-world scenarios. It is tested on video datasets, which demonstrate that the proposed approach provides promising accuracy if the frames are processed in real-time. This makes it particularly effective in self-driving cars, where decision-making has to be done as soon as it is possible based on the information that is received in the form of visual data. The paper serves a rather valuable intervention in the field by showing that traffic sign recognition in video is possible in real time, an improvement over merely recognizing static images. Their research focuses on integrating reliable feature extraction with effective low-power processing to satisfy the real-time requirement of autonomous driving applications.

Paper 3 - "Traffic Sign Classification with Dataset Augmentation and Convolutional Neural Network"

Some of the questions that the paper seeks to answer include how effective and efficient dataset augmentation in traffic sign classification is employing CNNs. Realizing the fact that rich and varied amount of data remains a key element in the successful training of deep learning models, the authors pay much attention to data enhancement of the current traffic sign dataset to increase the performance and versatility of the learned model. Therefore, rotation of images and variation in scale and colour are used to expand the dataset in order to ensure that the CNN model is exposed to more complicated visual scenarios.

CNN architecture proposed in this paper will be useful for identification and classification of the traffic signs belonging to different classes. In the paper, the authors increase the dataset through augmentation, which the authors show yields a higher classification accuracy and is useful in addressing misclassifications in different scenarios including change in angles, lighting and partial occlusion. The paper also underlines the necessity to have diverse datasets for CNN models' training since the augmentation data contribute to the model's enhanced resistance to real-life variations in traffic signs' appearances. In this work, the author demonstrates that dataset augmentation can be used in conjunction with CNN in order to enhance traffic sign classification capability of traffic signs. The findings also show that with the application of deep learning integrated with data augmentation, robust and accurate traffic sign recognition systems can be built which will be instrumental in the advancement of self-driving car systems.

Paper 4 - "Robust Real-Time Traffic Sign Detection Using a Probabilistic Fusion Method"

Here in this research paper published in 2013, the authors present a traffic sign detection approach that is more effective for real time traffic sign detection in the context of autonomous vehicles. The authors suggest a probabilistic fusion method which involves the use of multiple detection methods for the purposes of enhancing traffic sign

recognition in adverse scenarios including lighting conditions, blocking, and complex backgrounds. The other method is the fusion method, which combines the results of different models so that the system has high confidence on the probability of the detected result being correct. This approach is most helpful in complex driving situations where signs may either partially be hidden or one is called upon to read traffic signs when they are hard to see. Through integration it is possible to achieve recognition of the fact that the system is able to overcome the shortcomings of each individual detection method, thereby improving its overall effectiveness. This method is tested in real time situations and the results prove that the method is capable of controlling speed and accuracy associated with autonomous driving. Therefore, this paper enriches the existing knowledge by discussing the drawbacks of single-method detection systems as well as demonstrating that the integration of multiple probabilistic models improves the stability of traffic sign detection. This work is thus an important addition to the body of knowledge in improving the ability to create accurate, real-time recognition systems that work efficiently in the challenging environment upon which autonomous cars are deployed.

Paper 5 - "Multiple-Dataset Traffic Sign Classification with OneCNN"

Here they present a method to classify traffic signs with a single CNN trained on multiple data sets. The main problem solved in this work is the problem of merging data obtained from diverse sources with several types of traffic signs, differences in their visual quality and different labelling. To overcome this, the authors introduce the OneCNN model aimed at training the model on multiple datasets at once and therefore improving the model's generalization in the various environments and traffic sign types. The general architecture proposed by the OneCNN is designed to adapt to the inherent differences between datasets so that no retraining occurs on new data. The model was trained on several benchmark traffic sign datasets such as GTSRB and other datasets related to other regions and the results demonstrated decent accuracy and cross regional generalization capabilities.

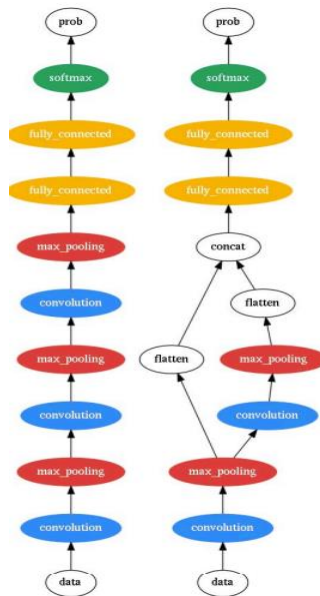


Figure 3:LR Logic diagram of the model

The above images represent the logic diagram of the model, and this provides a clear insight regarding the model development. That is why one of the major findings of this paper is the potential of OneCNN to preserve high classification rate while training on different data. This approach is much more practical compared to the creation of models tailored for various regions as is the case of traffic signs that differ from country to country in case of self-driving cars. The authors have produced work that will be beneficial to further research on multi-dataset training and its future enhancements to the performance of traffic sign recognition systems.

Paper 6 - "Issues and Future Directions in Traffic Classification"

Here the paper focuses on the difficulties and prospects of traffic classification in computer networks. Since there is a wide range of applications and services running over the network the issue of classification has become crucial and challenging due to which the authors have contributed to their paper. They debate the drawbacks of conventional approaches, namely port-based and payload-based classification, which fail to work due to the usage of encryption, tunnelling, and dynamic ports.

Here are the main problems discussed in the paper: there is a lack of methods for handling encrypted and obfuscated traffic, traffic classification becomes more difficult as new networks grow larger and more complex, and the approach for building real-time

traffic classifiers should be highly accurate. The authors stressed that machine learning and statistical techniques should be used to address these challenges as solutions. Machine learning can also use patterns in traffic behaviour and flow level statistics and get more elastic and stable classification kinds without having to do with the content of the packets.

Besides, the paper highlights the further development of traffic classification investigating possible trends, and points at the necessity of cooperation between network researchers and machine learning specialists. For their part, the authors believe that the traffic classification will be a vital tool to enhance the network security and quality of service as well as the management of resources as the network environment changes.

Paper 7 - "CNN Based Traffic Sign Classification Using Adam Optimizer"

Here, in this 2019 paper, the researchers developed a Convolutional Neural Network (CNN) model for traffic sign classification and employed the Adam optimizer to help improve the model.

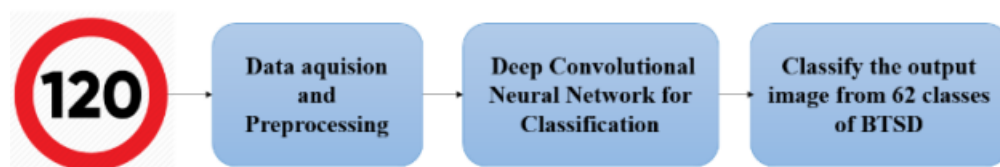


Figure 4:LR Planning process of the project

The study is centred on optimizing the CNN network such that it has the ability to classify traffic signs correctly and in the shortest time possible. The authors also take the time to illustrate its benefits from using the Adam optimizer where this is a learning algorithm that self-adjusts for learning rates for each parameter that makes it highly preferable in conditions with noisy gradients and non-stationary objectives.

The CNN architecture built in this paper is a sequential composition of a number of convolutional layers, pooling layers, and fully connected layers, which is the goal to learn the fine-grained features from traffic sign image. The Adam optimizer is very vital in the training procedure to achieve fast convergence while at the same time reaching the best

model solution. To support this, the authors test the performance of their model on traditional traffic sign datasets with classification accuracy rates being high.

The present paper is useful for the given field of research by applying the CNN architecture along with the Adam optimizer to classify the traffic signs effectively and efficiently. The findings indicate that this strategy can be useful in decision-making of real time traffic sign recognition applications for self-driving cars in which quick determination is crucial. The study also creates room for the future research in enhancing the performance of CNN particularly for real life applications.

Paper 8 - "A Practical Approach for Detection and Classification of Traffic Signs Using Convolutional Neural Networks"

Here the paper introduces how to find out the traffic signs and then classify them through Convolutional Neural Networks (CNNs). The authors stress on the development of powerful and effective traffic sign recognition systems for autonomous vehicles since the classification should be done in real time and with high accuracy. Their approach addresses both the detection and the classification sub-problems using CNNs in order to address the variation in sizes, shapes, and conditions of traffic signs.

Towards this end, the paper is organised in two parts. First, a region proposal technique is employed to localize the possible traffic sign locations in an image. Then the CNNs are used to identify the regions into different traffic sign categories. The CNNs are applied to the detected regions to categorise the regions into various traffic sign classes. The authors pay much attention to the architecture of CNN for matching both performance and complexity, which will be useful because in the real-life scenarios the resources are limited. The system is then tested on various traffic sign datasets where it performs fairly well for detection as well as classification of traffic signs.

It is noteworthy that the work of the researcher is useful for the actual application of CNNs in traffic sign recognition systems. What's more, their approach consists of region detection with CNN-based classification, which provides an effective solution and can be applied to AVs immediately. The paper also discusses difficulties of model application in the real conditions mentioning obstructions and changes in lighting conditions and indicates some directions for future investigations.

Paper 9 - "Traffic-Sign Detection and Classification in the Wild"

In this 2016 paper, the subject of the study revolves on the traffic sign detection and classification in natural or 'real world' environment which goes under the term of 'traffic sign detection in the wild'. Unlike the previous studies where the environment is purely controlled with clear distinguishable traffic signs with variations in lighting conditions, partially occluded and on complex backgrounds. In order to overcome these challenges, the authors suggest a system that integrates deep learning approaches with superior object detection frameworks.

Their method for Traffic Signs detection entails using the region-based Convolutional Neural Networks (R-CNNs) to identify signs within an image then classification of the detected signs. The system has been developed to withstand such factors of variability and uncertainty that are found in real life such as partial occlusion and low visibility. To evaluate the effectiveness of their approach, the authors apply it to complex traffic sign data sets which is done with high accuracy and combined with an analysis of their technique, the authors show that the proposed approach outperforms existing solutions in various scenarios.

The work done by the Researcher is crucial as it takes traffic sign recognition outside the laboratory environment as a testbed. They have complemented their findings with the practical application to show how their deep learning models may be applied on self-driving cars where traffic sign detection is critical in order to avoid fatal mistakes. **Paper**

10 - "Traffic Sign Classification Using Deep Neural Network"

This paper presents a deep neural network (DNN) approach to traffic sign classification, aimed at improving the accuracy and reliability of traffic sign recognition systems in autonomous vehicles.

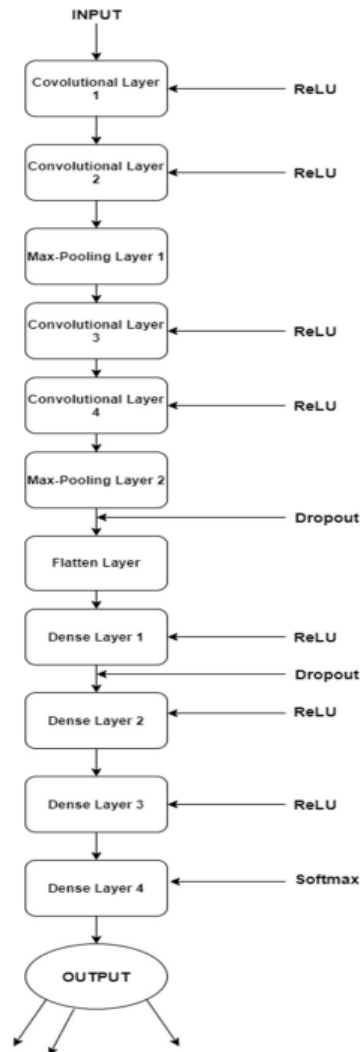


Figure 5:LR Logic diagram

The authors proposed deep learning architecture for traffic sign images to cope up with different shape, colour & size of signs, and hurdles like occlusion and severe weather affects visibility of the signs.

The main idea of the proposed approach is the extension of the DNN model through incorporating several layers which serve to filter and extract relevant features from the images in order to facilitate properly categorization of traffic sign images. As for any machine learning algorithm, the authors try and tweak various hyperparameters as well as employ several optimization algorithms. Also, they point out that when training the model, it is essential to use several datasets in order to enhance the model's ability to generalize on real-world data.

By the help of testing the accuracy rates of the model are high on standard traffic signs databases, thus proving its efficiency in identifying traffic signs under various circumstances. The paper also shows the ability of deep neural networks in enhancing traffic sign recognition technologies that can add to the more extensive area of autonomous vehicles and smart transportation systems.

These findings stress the importance of a deep Learning in traffic sign classification and suggest that DNNs can result in substantial performance enhancements as well as improved accuracy compared to verified machine learning approaches.

3. Project Description

3.1 System Requirements

To ensure the smooth training and testing of the traffic sign recognition system, the project necessitated both local PC hardware and cloud-based resources for large-scale model training. Below are the system requirements:

PC Requirements:

- **Processor:** Intel Core i5 or higher
- **RAM:** 16 GB or more
- **GPU:** NVIDIA GeForce GTX 4070 or higher (with at least 8 GB of VRAM)
- **Storage:** 500 GB SSD for fast read/write access

Cloud Requirements:

- **Cloud Provider:** Google Collab Pro
- **Compute Instances:** High-performance GPU instances (like in collab TPU.v2)
- **RAM:** 170 GB for larger datasets and complex models
- **Storage:** Access to cloud storage like Google Drive, as it be storage for handling large datasets and saving model checkpoints

The cloud computing's resources were utilized for tasks that required more processing power like hyperparameter tuning, data augmentation and training with big data sets. These tasks were more appropriate for cloud-based GPU instances as they would allow the load to be processed efficiently.

3.2 Libraries Used

The following libraries were utilized throughout the project, each serving a specific purpose in the development and deployment of the traffic sign recognition system:

- **TensorFlow/Keras:** TensorFlow and particularly Keras API, were the main platform utilized throughout the development, training and assessment of the developed deep learning models. Keras has a simple API for developing the complicated deep learning models, it also supports transferring learning, CNN and many other complex techniques like data augmentation etc.
- **OpenCV:** For image preprocessing like image resizing, image cropping and data augmentation the OpenCV library was employed. This is beneficial in handling image data and applying transformation on the image which enhances the model training.
- **Scikit-learn:** Scikit-learn was helpful in instance segmentation, scoring, and computing other machine learning metrics such as accuracy and precision. It also helped with procedures for scaling and normalization of data.
- **NumPy and Pandas:** Data Management and manipulation were handled using NumPy and Pandas. The array computations were majorly from NumPy, and handling and cleaning of the datasets were from Pandas.
- **Matplotlib/Seaborn:** These visualization libraries were employed to generate plots and graphs for EDA, monitoring model performance, and results presentation. They assisted in class distributions, training curves, and model metrics to evaluate the model.
- **Augmentations:** This image augmentation library was employed to perform more intricate augmentations including the random manipulation of brightness and contrast as well as applying zoom. Specifically, it was especially helpful in

handling issues related to dataset skewness and enhancing the model's ability to generalize on unseen data.

These libraries collectively enabled the implementation of complex deep learning techniques, model training, and performance evaluation.

3.3 Techniques Used

This project used various advanced methods in machine learning and deep learning to create the efficient traffic sign recognition system. Below is a brief description of the key techniques used:

- **Neural Networks (NN):**

A neural network (NN) is a mathematical model which is designed based on the structure of the human brain and comprises interrelated nodes or neurons layered. NNs have the ability to predict the model based on the data given to them and is used in machine learning for classification and regression. In this project, neural networks were the base models on which the traffic sign recognition models were developed from.

- **Convolutional Neural Networks (CNNs):**

CNNs are a type of deep neural network that falls under the class of narrowly defined deep learning models used for handling structured grids for instance images. CNNs apply convolution layers which help the networks to learn the hierarchy of features from the input image, in an automatic manner. This makes them suitable for solving image recognition problems such as detecting traffic signs. The CNN architectures that were used in this project had many convolutional layers, pooling layers, and fully connected layers that were used to extract and classify features in the traffic sign images.

- **Transfer Learning:**

Transfer learning is the practice of reusing a pretrained model, which usually has been trained on one or more large datasets such as ImageNet, for a new task, such as traffic sign recognition. In this project, transfer learning was used using MobileNetV2 model architecture. This model was initialized with pre-trained

weights, and then it was trained again with the traffic sign datasets to be more precise especially when dealing with limited data sets.

- **Data Augmentation:**

Data augmentation is the process of increasing the quantity of data in order to obtain a new data set from the initial data set through the use of small random changes to inputs, which may include rotating, flipping, zooming or changing colour of images used in the training process. In this project, data augmentation was used to combat dataset imbalance and improve the model's generalization performance.

- **Hyperparameter Tuning:**

Hyperparameters are the parameters controlling the learning process of the model, and they include the learning rates, batch size and the number of layers. While performing the hyperparameter tuning, popular methods like the grid search as well as random search were used to determine the best values which brought out the highest possible performance from the model.

- **Model Versioning:**

Different CNN models were built and trained beginning from basic structures and then moving to complex designs. The final selected model for the German dataset incorporated a different and higher number of layers CNN architecture after many trials of augmenting, balancing and selection of the best layers among them.

4. Methodology

4.1 Planned Methodology & Data Sources

4.1.1 Planned Steps to be Conducted.

The research methodology for this project is designed based on the systematic approach of learning which has been used for the formulation of traffic sign recognition using Convolutional Neural Networks (CNNs). The following steps outline the planned process:

1. **Data Collection and Preparation:**

- **Dataset Acquisition:** The first prerequisite includes finding the adequate traffic sign datasets that will be used in learning. Three major datasets were chosen, each representing different regions: Such databases include the German Traffic Sign Recognition Benchmark (GTSRB), the Indian Traffic Signs Dataset, and the Asian Traffic Signs Dataset.
- **Data Preprocessing:** After these datasets are obtained then the images are pre-processed. This entails scaling the images down to a smaller size (32x32 pixel) and Normalization the pixel concentration values to the range of [0 1] and splitting the training data set into training, validation & testing data already exists.

2. **Exploratory Data Analysis (EDA):**

- EDA is carried out to analyze the structure and distribution of the datasets. Overviews are created to analyze the distribution of classes, the quality of images, and other attributes that may affect model performance.

3. **Model Selection and Design:**

- **CNN Architecture Design:** As a result of EDA, several CNN architectures are developed is as follow: These models will have several convolutional

layers with pooling layers that reduce the spatial dimension of the data, fully connected layers, and dropout layers to minimize overfitting.

- **Transfer Learning:** In addition, proposal of developing CNN models from scratch, for data set which aren't able to use CNN effectively is the case where transfer learning techniques comes. Here pre-trained models like MobileNetV2 are explored.

4. Training and Hyperparameter Tuning:

- **Model Training:** All the CNN models are applied on the pre-processed datasets using the training methodologies such as stochastic gradient descent and backpropagation (In short Neural Network). This kind of parameters includes learning rate, batch size, number of epochs, among others that are tuned to make a model perform better.
- **Hyperparameter Tuning:** The last step is hyperparameters tuning for each model, where a grid search or random search is performed to achieve higher accuracy and better generalization.

5. Data Augmentation:

- To avoid overfitting, data augmentation strategies are used during the training of the model. These include random zoom, brightness shift, grey scale, and contrast shift to mimic variation in the training data set. Because DataImageGenerator is used it randomly selects one of them mentioned augmentation and between the scale it applies to each image which makes nearly all images unique almost each epoch.

6. Model Evaluation:

- **Evaluation of 3 models:** The models are measured based on values like accuracy on the validation data and test data among others. This step helps prevent the models from overfitting and provides an estimate of how well generalization of models will be on unseen data.

- **Inter-Data set testing:** Run each of the 3 models on the test set of the others in order to understand how the modelling is impacted by new geographical Traffic signs.
- **Real-Time Testing:** Real time video feed data set is also used to evaluate the performance of the final trained model in realistic settings. This is important in establishing the viability of the model in real life driving scenarios.

7. Comparison and Analysis:

- The findings of the different models are considered and evaluated with regards to their advantages and limitations. The results are then presented according to the research objectives with the aim of enhancing traffic sign recognition in various areas and conditions.
- More exploration is done on the final model like Which type of signs after grouping have higher or lower accuracy and compare to the nod of images in a group.

4.2 Data Sources

As stated earlier, the effectiveness of this research relies on the nature and variety of the datasets utilized for training and testing. Three datasets were selected to ensure a comprehensive analysis of traffic sign recognition across different regions and conditions.

4.2.1 Asian Based Traffic Signs Dataset

The Asian Traffic Signs Dataset is obtained from Kaggle, and it is comprised of approximately 6,000 images together with 52 classes of traffic signs. Most of the signs you will encounter in the dataset are Chinese traffic signs with specific regional differences due to different rules in Asia. This is made up of regulatory signs, warning signs, and Informative signs and many of these signs are usually posted in many languages due to the diverse linguistic landscape in the region.

Dataset Composition

The dataset contains training and testing data with the images that are already categorized making it a supervised learning set. These signs are captured under scenario such as under different lighting and/or weather condition, therefore, it is preferred for training Convolutional Neural Network (CNNs). For the Traffic Sign Recognition model, it contains general traffic signs such as speed limits signs, stop signs, and signs specific to the regions only, making it quite challenging.

Sample Images

Below are two sample images from the dataset:

1. **Unique Traffic Sign:** This image represents a traffic sign unique to this dataset (Asian Traffic dataset).



Figure 6: DS1 traffic signal ahead warning sign

2. **Common Traffic Sign:** This image shows a common traffic sign, such as a "Speed Limit 30" sign, which appears in all 3 traffic sign datasets and nearly identical globally.



Figure 7: DS1 Speed limit 30

The dataset does not give the exact location of the region of interest in the images but rather assign each image to a class. These images are grouped in folder and each folder

is having only the images of one class only. Also, a CSV file that contains the class numbers and corresponding sign names exist. At times, some of the classes are not named in the label file as expected. Instead of dropping these images and decreasing the number of data, the actual names of these classes were discovered, and then corrected alongside its label file.

Table with updated signs from unknown:

ClassId	Name	Updated Name
9	Unknown7	Yield Sign
33	Unknown6	Stop Sign
40	Unknown1	Steep Hill Warning Signs
41	Unknown2	Steep Hill Warning Signs
42	Unknown3	Slow Down
45	Unknown4	village or residential area ahead
49	Unknown5	Traffic Light Warning Sign

4.2.2 German Traffic Sign Recognition Benchmark (GTSRB)

One of the largest and well-known datasets is the German Traffic Sign Recognition Benchmark (GTSRB). The data used in the current study is obtained from Kaggle and consists of more than fifty thousand images of German traffic signs classified into 43 classes. Therefore, this dataset is helpful due to the nature and number of images provided and such factors as lighting or angles and even the degrees of occlusion. Additionally, this data set has been used few of papers mentioned above in literature review.

Dataset Composition

The German Traffic Sign Recognition Benchmark (GTSRB) is one of the largest datasets that has been widely used in the traffic sign recognition field. The given data set is obtained from Kaggle which involves more than fifty thousand images of German traffic signs divided into 43 sign classes. This dataset is so useful because of the variety of

images included and the conditions under which the images were taken such as lighting or angles and even degrees of occlusion.

Sample Images

Below are two sample images from the dataset:

1. **Unique Traffic Sign:** This image represents a traffic sign that is unique to the GTSRB dataset, illustrating the dataset's focus on region-specific signs.



Figure 8: priority road ahead

2. **Common Traffic Sign:** This image shows a common traffic sign, such as a "Speed Limit 30" sign, which appears in all 3 traffic sign datasets and nearly identical globally.



Figure 9:DS2 Speed limit 30.

4.2.3 Indian Traffic Signs Dataset

The Indian Traffic Signs Dataset available at Kaggle is a rich set of traffic sign images specifically collected from different areas of India. This dataset contains more than 20,000 images divided into 85 different categories/classes and is therefore the most expansive dataset in terms of the number of classes. The dataset covers Indian traffic signs which are quite different from Western and other Asian traffic signs due to multilingualism and diverse Indian roads. Many of these signs are imperative for the functioning of self-driving cars in India where drivers come across both formats of signs, most of which include symbols and words in multiple languages.

Dataset Composition

The dataset is divided into separate folders for training and testing, with each image pre-labelled to help supervised learning tasks. The traffic signs range from standard regulatory signs, such as stop and speed limit signs, to more region-specific signs that reflect the country's diverse linguistic and cultural landscape. These unique signs often feature text in local languages, which adds complexity to the traffic sign recognition task.

The Indian Traffic Signs Dataset is particularly valuable for testing the adaptability of machine learning models to non-Western traffic environments. With 85 distinct classes, this dataset offers the widest range of traffic sign types in this research, making it a comprehensive resource for developing models that can generalize well across varied sign formats. The diversity of the signs in terms of both visual appearance and content ensures that models trained on this dataset can handle a variety of real-world scenarios that are typical in Indian traffic systems.

Sample Images

Below are two sample images from the dataset:

1. **Unique Traffic Sign:** This image represents a traffic sign that is unique to the Indian dataset, displaying the regional specificity of the signs.



Figure 10: Cattles ahead

2. **Common Traffic Sign:** This image shows a common traffic sign, such as a "Speed Limit 30" sign, which appears in all 3 traffic sign datasets and nearly identical globally.



Figure 11:DS3 Speed limit 30

This dataset is useful in training models that are capable of identifying traffic signs commonly found in Indian road environment. In this way, model can train on the manifold of Indian traffic signs and help them to cope with linguistic and visual differences which is useful for training applicative models for real self-driving car systems in India.

4.3 Class Distributions

Understanding the distribution of classes within each dataset is crucial for identifying potential challenges in model training, such as class imbalances.

Class Distribution in the Asian Traffic Signs Dataset

The distribution of the classes in the Asian Traffic Signs Dataset highly skewed, that is, some classes have many images, and some have few images. In certain traffic sign types, it is obvious that there are many images more than 500 while in other types there are images less than 100. This skewed distribution shows that the model may develop a preference for identifying the majority class, which may have a negative impact on the signs that are not as frequently used. Such imbalances will need to be balanced through measures such as data augmentation or class weighting to enable the model effectively to generalize to all classes.

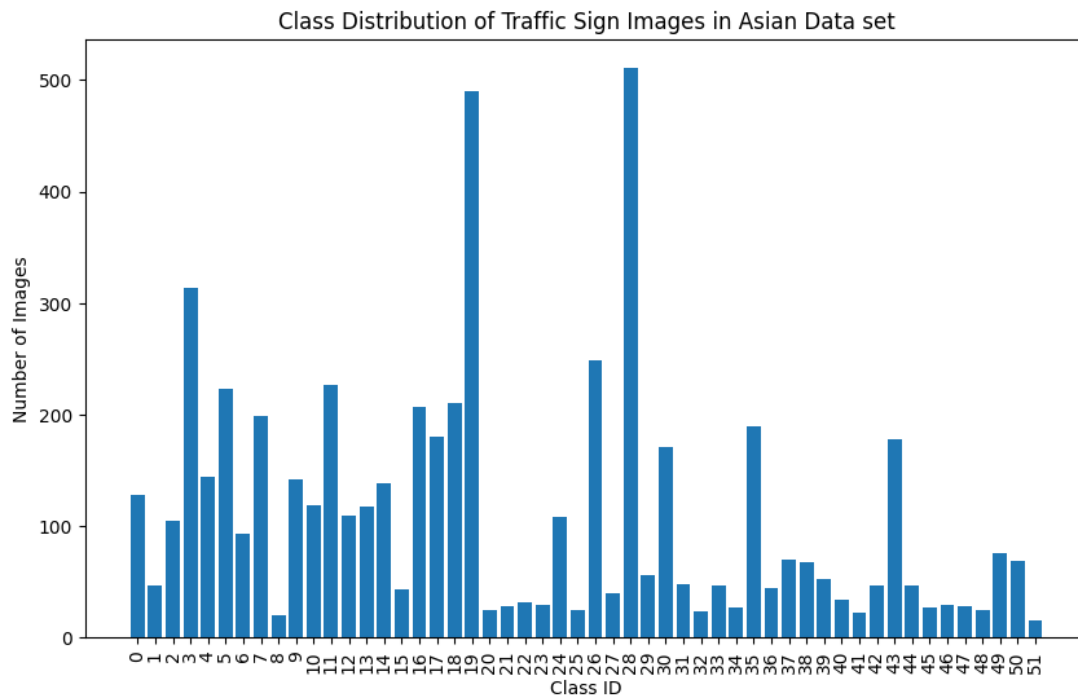


Figure 12: DS1 images distribution by class

Class Distribution in the German Traffic Sign Recognition Benchmark (GTSRB)

Across the 43 classes, the GTSRB dataset also demonstrates significant imbalance similar to the observed with the Asian dataset. For some traffic sign classes, there are more than 2,000 images, whereas for others, there are fewer than 500 images. This imbalance will more so impact the model’s capability of accurately predicting signs from the least occurring classes. This distribution pattern is alike to the one identified in the Asian dataset boosting awareness of techniques such as data augmentation and balanced class sampling for covering every aspect possible during the model training process.

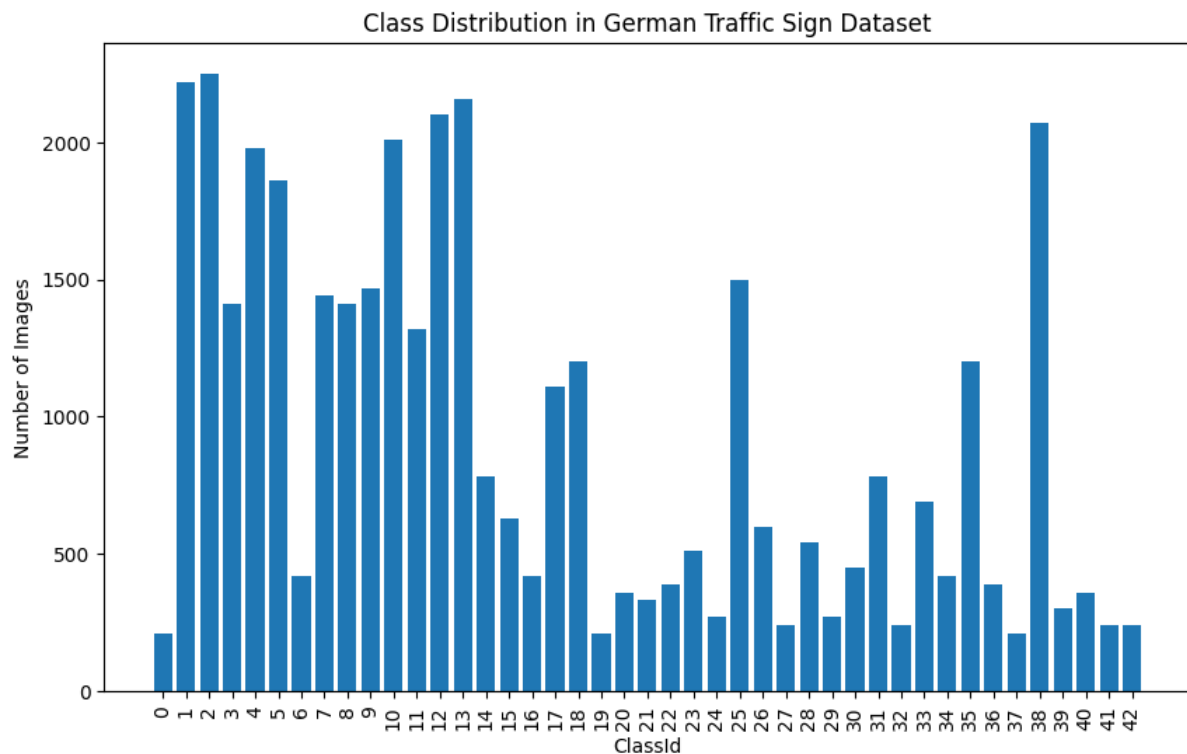
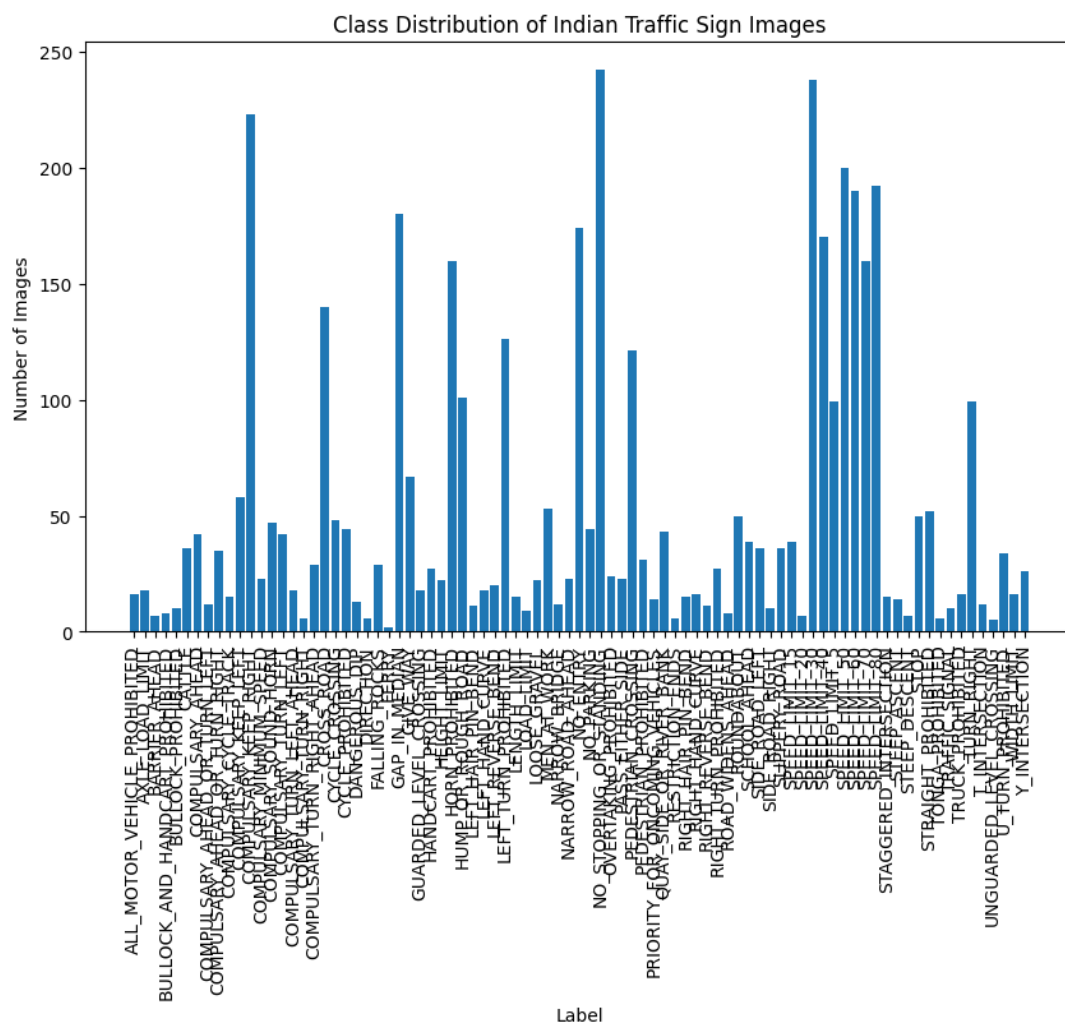


Figure 13:DS2 images distribution by class

Class Distribution in the Indian Traffic Signs Dataset

Indian Traffic Signs Data set contains 85 different classes and again in this data set also the classes are not equally distributed. Most of the labels possess more than 200 pictures, while there are some with only 20 pictures. Such imbalance in classes as seen in the previous datasets can easily result in the model being biased to the classes that are most frequent. Such biases are inevitable when creating models for such diverse and complex dataset hence needs special attention during the model training and possibly establishing methods that oversample or creating customized loss functions to increase the cost of misclassification on the minority classes.



4.4.2 Model Architecture

The final model architecture used transfer learning with **MobileNetV2** as the base model. This approach was selected after traditional CNN architectures failed to perform effectively, particularly with the complex Indian Traffic Signs Dataset.

- **MobileNetV2 Base:** The base model, pre-trained on ImageNet, was frozen to retain the learned features.
- **Custom Layers:** Additional layers were added, including Global Average Pooling, Dense layers with ReLU activation, and Dropout for regularization.
- **Output Layer:** The final layer used SoftMax activation to output class probabilities corresponding to the number of traffic sign categories in each dataset.

This architecture provided the necessary flexibility and generalization capability, especially when handling the diverse traffic signs in the Indian dataset.

4.4.3 Data Augmentation

In the initial phases of training, various data augmentation techniques were tested, including brightness shifts and grayscale conversion. However, these augmentations caused the model to fail in learning effectively. After extensive experimentation (trial & error), only random zoom was retained as the augmentation technique, with a zoom range set between 0.8 and 1.2. This adjustment allowed the model to benefit from augmentation while still learning meaningful features from the data.



Figure 15: All planned augmentation, image from DS3

By narrowing the focus to random zoom, the model was able to handle variations in the distance of traffic signs from the point of view without introducing unnecessary complexity from other augmentations that hindered learning.

4.4.4 Training and Hyperparameter Tuning

The model was optimised using Adam Optimizer and categorical cross-entropy function. The hyperparameters, including the learning rate and batch size, were adequately adjusted to ensure maximum performance by the model.

- **Learning Rate:** Various learning rates were tested, with a final selection that balanced training speed and model convergence.
- **Batch Size:** A moderate batch size was used to ensure training stability without overwhelming memory resources.
- **Early Stopping:** Early stopping was implemented to prevent overfitting, halting training once validation loss stopped improving.

Finally, the complex & resource demanding part of training & test was performed on Google Collab Pro since the code was able to leverage GPU acceleration due to the demands of the current training and test data set used. Some modifications were done depending on available resource when building larger models and also when training for longer periods of time.

4.4.5 Model Evaluation

The final model performance was assessed by utilizing standard accuracy measures as applied to the validation and test data sets. Instead of the inter-dataset testing as a validation of the model, the mixed set testing was carried out in order to select the final model for the project because of the problems arising from mapping the labels from one dataset to another.

- **Mixed Set Testing:** A combined dataset was created by merging samples from all three datasets (German, Indian, and Asian Traffic Signs). This mixed set allowed for a more comprehensive evaluation of the model's ability to generalize across several types of traffic signs from various regions. By testing on this mixed set, the final model was selected based on its performance across diverse sign categories.

4.4.6 Real-Time Testing

Apart from the evaluation of the models on static databases, the final selected model was also tested in a real time environment for the evaluation on video feeds. In this test, the model was fed with videos and on each frame an estimate was made on the tenth frame. The real time scenario resembled the dynamic situations that the self-driving car faces daily, including variation in light, view angle, and speed of the moving car.

4.5 Logic Diagram of the Final Model

4.5.1 Logic Diagram

The below figure depicts the final architecture of the Convolutional Neural Network (CNN) model applied for the traffic sign classification. This flow diagram shows how the information is processed from input layer into the output layer, indicating what is done at each layer of the communicating network.

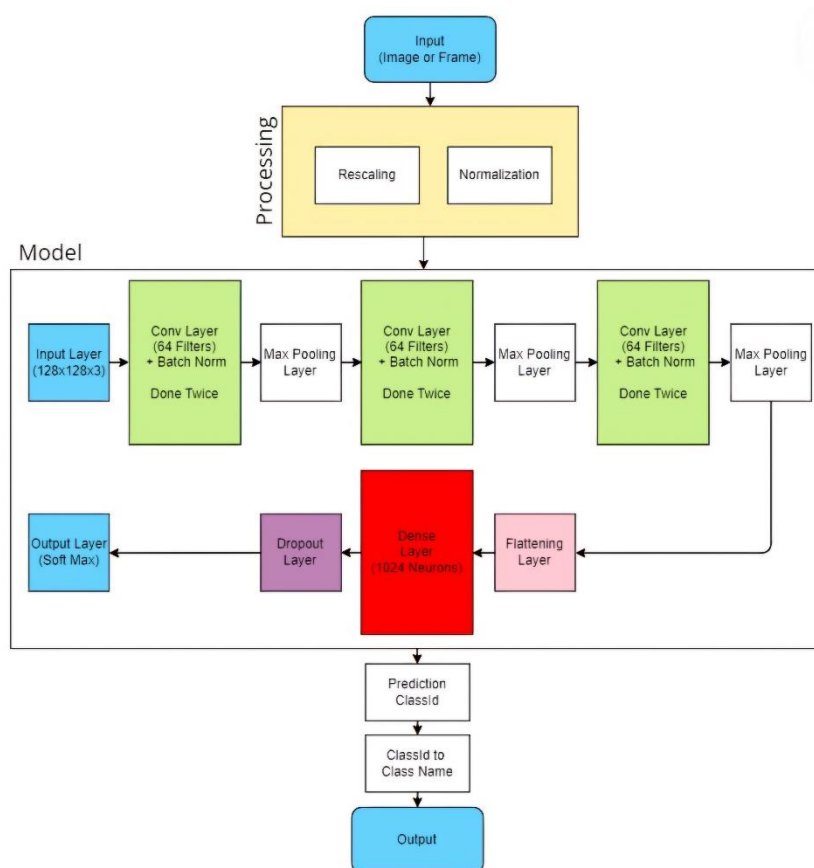


Figure 16: Project Logic Diagram

4.5.2 Explanation of the Logic Diagram

As seen from the logic diagram, the final model is composed of several convolutional layers, batch normalization, max pooling, and dense layers. Here's a brief breakdown of each section of the model as illustrated in the diagram:

1. **Input and Preprocessing:** The model first layer is the input layer takes in the images formatted to 128 x 128 pixels with three channels (RGB). Before feeding the images to the model, the images are pre-processed to make the data most suitable for learning these procedures include scaling as well as normalization.
2. **Convolutional Layers and Batch Normalization:** The main component of the model is made up of several convolutional layers, and each of them has been tempered with batch normalization. These layers then build on each other and extract increasingly higher-level features from the input images such as edges, textures and even more complex patterns. Batch normalization aids the training process to become more stable and faster, as it Favors normalization of the activations of the previous layers.
3. **Max-Pooling Layers:** Following the convolutional layers, the max-pooling layers are used in order to reduce the size of the feature maps (down sampling). This means that the spatial dimensions of the data are reduced in a way that enables the model to consider only the most relevant features while at the same time preventing model's computational complexity.
4. **Flattening and Fully Connected Layers:** The feature maps are then flattened into 1D vector which is then passed to fully connected (dense) layer of 1024 neurons. This layer plays the role of the decision-making component of the model where all the features extracted are used to carry out the classification.
5. **Dropout Layer:** To avoid overfitting, a dropout layer is included before the output layer. This layer randomly removes some of the neurons during training so that the model is fine-tuned to perform better when presented with an unknown data set.

6. **Output Layer:** The final layer is a SoftMax output layer that predicts the probability of each traffic sign class. The highest probability is selected as the predicted ClassId.
7. **Post-Processing:** After the model generates or selects class ID, it is mapped to the corresponding traffic sign name, providing the final classification result.

5. Results

5.1 Overview

This section presents the performance of models trained on three different traffic sign datasets: Asian in origin, German and Indian type. The outcomes of these datasets are then trailed by mixed datasets for testing, which was applied to assess the generality of the models all through regions. The final decision on which of the models to use in the subsequent procedures of augmentation, balancing and hyperparameter tuning and from which dataset they should be derived was made based on these results with the final decision being to use the model trained on the German dataset for the said subsequent procedures to create the final model.

5.2 Performance on the Asian Dataset

The model trained on the Asian dataset (Model 1) achieved a strong test accuracy of **92.17%**. The following graph summarizes the training and validation accuracy:

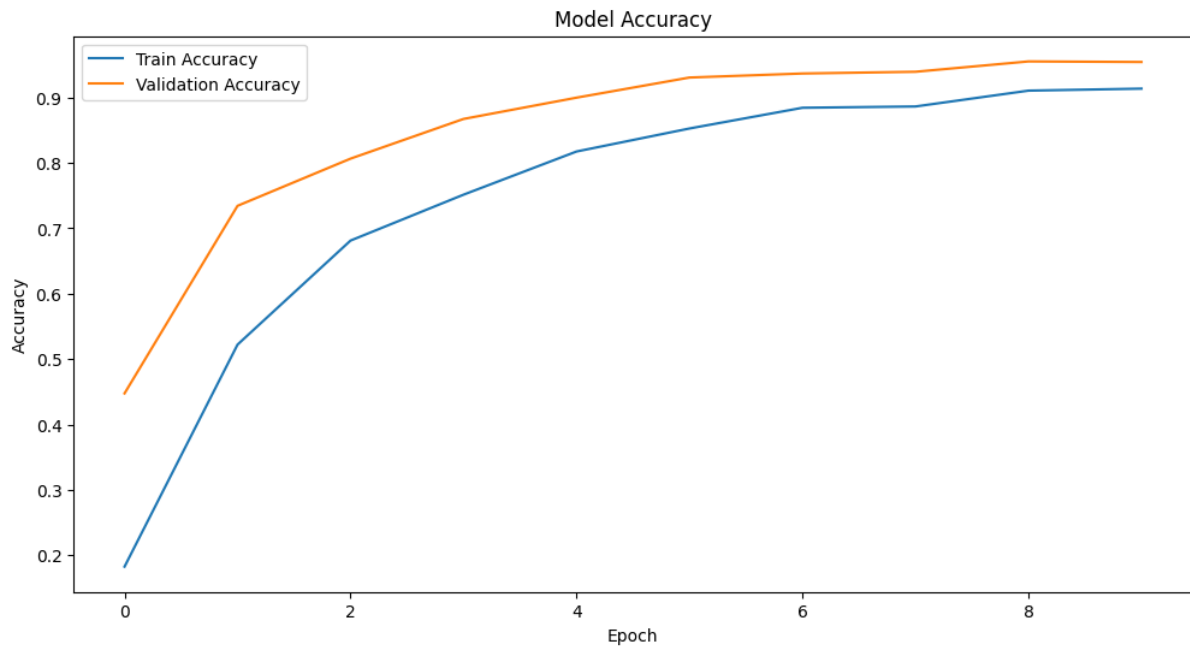


Figure 17: Model 1 Progress

The model achieved high test accuracy (92.17%), demonstrating effective learning on the given data set. Further enhancements, such as augmentation or balancing, were not applied as the final model, has to be chosen yet, however the model performed well on this dataset.

5.3 Performance on the German Dataset

The model trained on the German Traffic Sign dataset (Model 2) displayed excellent generalization, achieving a test accuracy of **99.49%**. The German dataset had the most images among all datasets about 60K, hence it shows contributing to the model's strong feature learning. Below is the performance summary:

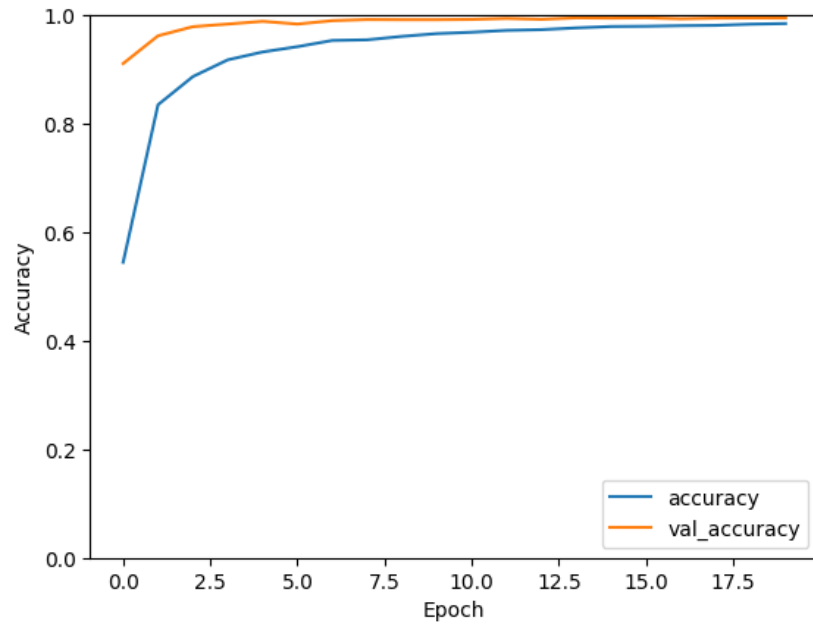


Figure 18: Model 2 Progress

The model performed exceptionally well, achieving a test accuracy of 99.49%. This performance can be attributed to the diverse and large dataset, which provided more training data for the model to learn robust features.

5.4 Performance on the Indian Dataset (DS3)

The model trained on the Indian dataset (Model 3) achieved a training accuracy of **97.28%**. Although the dataset presented challenges such as diversity and class imbalance, the model was able to handle these effectively:

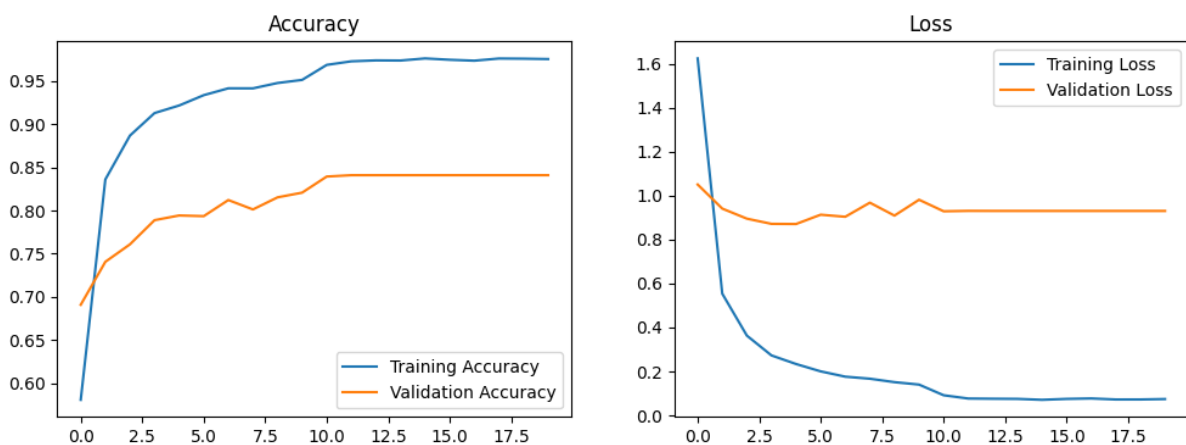


Figure 19: Model 3 Progress

Observations: Despite the challenges posed by the dataset, the model performed well, achieving a test accuracy of 84.08%. Similar to Indian data set's trained model further enhancements, such as augmentation or balancing, were not applied as the final model, has to be chosen yet, however the model performed well on this dataset.

5.5 Mixed Dataset Testing

In order to check the generalization ability of the models across different regions, mixed dataset test was performed. The findings suggest that the Model 2 which was trained on the German dataset gave the highest accuracy of 43.66%. This outcome was important in choosing the German dataset model for the further fine-tuning and finalizing step.

Mixed Test Results Table:

Model	Accuracy
Model 1 (Asian)	12.68%
Model 2 (German)	43.66%
Model 3 (Indian)	8.45%

Observations: Model 2, trained on the German dataset, demonstrated the highest generalization across different regions. The substantial number of images in the German dataset contributed to this model's success in learning a diverse set of features. Based on this impressive performance, Model 2 was selected for further refinement.

5.6 Final Model Selection and Refinement (German Dataset - DS2)

After selecting Model 2 (German dataset) as the best-performing model, further refinements were made through **augmentation, balancing, and hyperparameter tuning**. The training graph below shows the performance of the final model across 18 epochs, illustrating the accuracy and loss during training and validation phases.

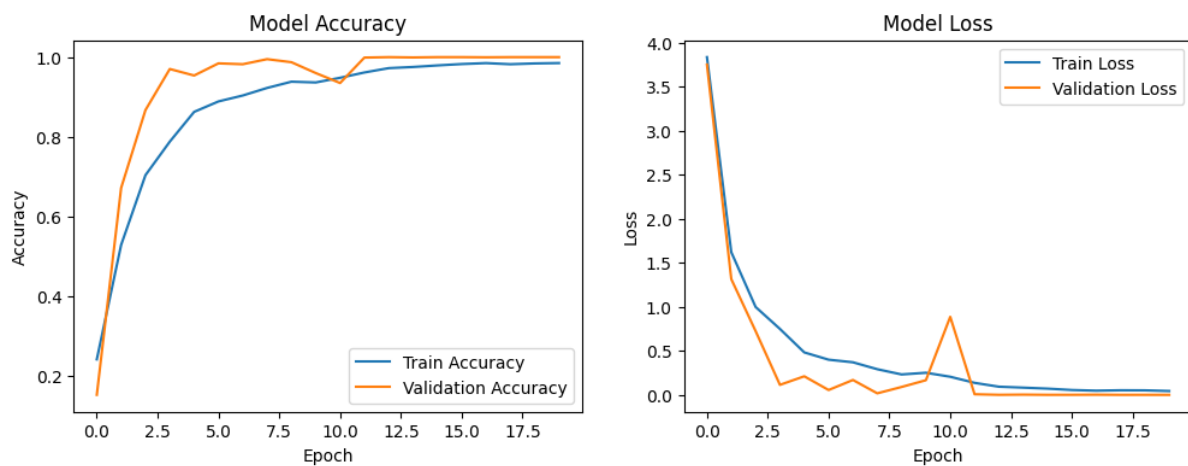


Figure 20: Final Model Progress

The final model showed a steady increase in training and validation accuracy, reaching near-perfect accuracy by the 10th epoch. The model's loss decreased consistently, indicating strong convergence. After hyperparameter tuning and applying augmentation and balancing techniques, the model achieved 100% training accuracy and 84.41% test accuracy.

Model Comparison Table:

Model	Training Accuracy	Test Accuracy
Model 2 Initial	99.67%	99.49%
Model 2 Hypertuned	99.68%	83.22%
Model 2 with Augmentation	52.17%	51.31%
Model 2 Balanced, Augmented & Tuned	100%	84.41%

The initial Model 2 performed exceptionally well, with 99.67% training accuracy and 99.49% test accuracy. However, the final refinements, including augmentation, balancing, and hyperparameter tuning, ensured that the model was balanced and generalizing well, ultimately achieving 100% training accuracy and 84.41% test accuracy.

5.7 Real-Time Prediction Testing

As a decisive step of model validation, the final selected model was run on actual video feeds to determine how well it performs in real-time settings. This real time testing mimicked real-world environment experienced in the Av system where light changes, view angles, and relative speed of the vehicle also form part of the environment for modelling. The video was processed with the model making predictions on every 10th frame to also ensure there was a balance between the number of predictions made by the model and the speed in which it made them while being resource efficient.

5.7.1 Real-Time Performance Observations

The real-time testing revealed several key insights into the model's ability to handle dynamic environments:

1. **Inconsistent Predictions:** While the model successfully handled static datasets, it failed to perform consistently in real-time conditions. For instance, the model provided different classifications of similar traffic signs in consecutive frames. This can be caused by minor differences in visualization for the object, for instance, in lighting conditions, position of the object relative to the camera, and the blur between frames.
2. **Environmental Challenges:** Additional factors which are related to the actual environment of vehicle motion such as snow, road work signs, and complex background increased the number of false predictions of the model. As illustrated in the following image, the snow in the background posed a problem by increasing the rate of false classifications and was also inconsistent with the results of the prediction.



Figure 21: Inconsistent prediction from video frame

It is clearly visible in above image that model exhibited inconsistency, classifying this scene with both "Road Work" and "Speed Limit 70" signs due to the snow and complex environment.

6. Discussion

6.1 General Overview of Results

The outcome deduced from the analysis of separate set of data proves significant in understanding the effectiveness and constraint of the learned models. Model 2, which was trained on the German dataset, again proved to be the most effective in both the static and real-time analyses. The mixed dataset also provided additional insight and emphasized that the current model already possesses great generalization potential, which is why it was selected for refinement and as the final model.

The test accuracy achieved in the final model is approximately 84.41%, which is reasonable to avoid overfitting on the training data while at the same time ensuring that our model is not too generic. This is a favourable outcome given the diversity of the traffic signs, lighting conditions, and environmental conditions arising from the real-world datasets. As for the limitations, the results suggest that further advancements are required to sustain the performance of the presented model under dynamic conditions in real-time setup.

6.2 Comparative Performance Across Datasets

- **Asian Dataset (Model 1):**

The model trained on the Asian dataset had a satisfactory performance with test accuracy at 92.17%. But this performance did not generalize well to the mixed dataset test where it recorded a mere percent accuracy of 12.68%. They concluded that, although the model was highly effective in training on that dataset, it was less effective in training on traffic signs in different regions. The given set of data was smaller, and there could be the overfitting of features of this or that region which also could have influenced this limited generalization.
- **German Dataset (Model 2):**

The second model that was trained on the German dataset had the highest test accuracy of 99.49% and higher accuracy in the mixed dataset test with the accuracy level of 43.66%. The model's ability to generalize across different traffic signs from various regions demonstrates the importance of dataset diversity and size in training robust models. From the outcome of this work, it can be deduced that more diversified and larger datasets are appropriate to implement accurate traffic sign recognition models. This model augmented, balanced, and optimized with hyperparameters that fine-tuned it for accuracy and generalization.
- **Indian Dataset (Model 3):**

Model 3 achieved a test accuracy of **84.08%**, indicating effective learning within its dataset despite the challenges of diversity and class imbalance. However, like in Model 1, it was ineffective to generalize within the mixed dataset test, resulting with an accuracy of 8.45 percent. This inferior performance highlights the challenging nature of Indian traffic signs which includes multilingual and region-specific signs, therefore making it almost impossible to generalize to other datasets.

6.3 Interpretation of Results by Traffic Sign Group

The performance of the final model can be better understood by analysing its accuracy across different traffic sign groups.

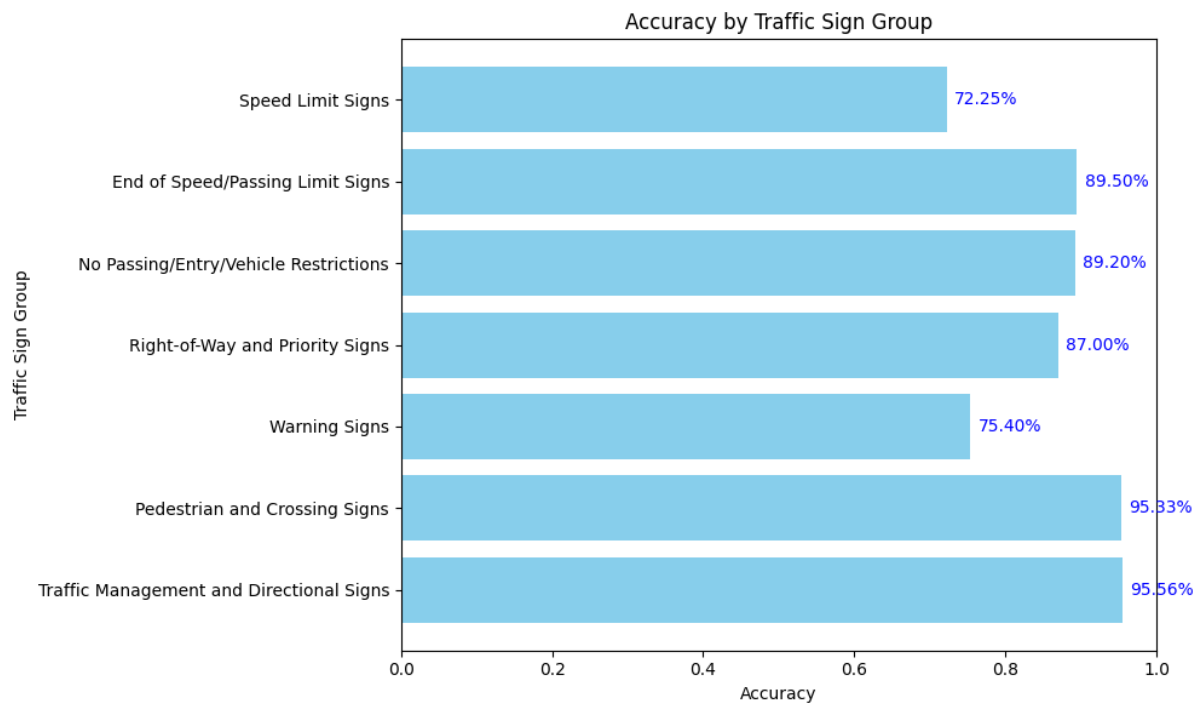


Figure 22: Comparison of accuracy by group.

As mentioned earlier that grouping of traffic signs into categories based on their use, function reveals how the model performed in recognizing signs that share similar patterns and characteristics.

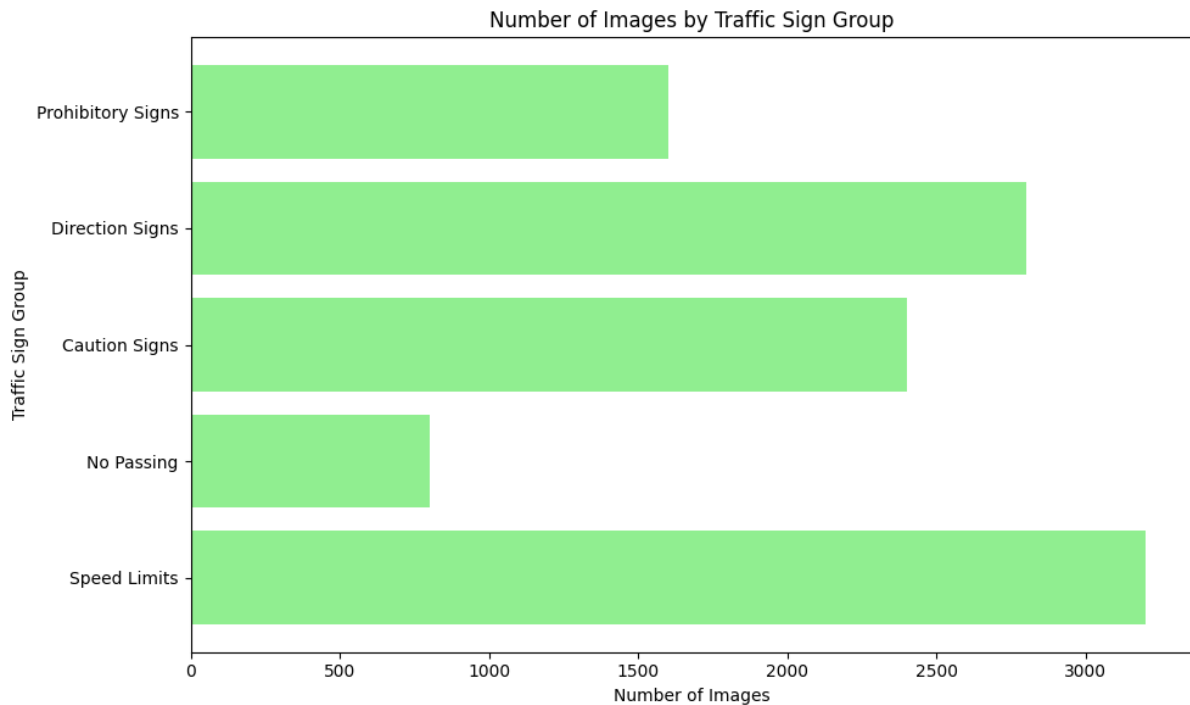


Figure 23: Comparison of No. of images per group.

- Speed Limit Signs (Group 1):** on the speed limit signs the model achieved an accuracy rate of 72.25%. This category posed some difficulty in the approach because most of the speed limit signs are very much alike such that the only variation is the figures displayed. It is for this reason that the model struggled to discern the differences between the numerous speed limit images in the dataset due to their richness and differences in small details. This indicates that even with a relatively large amount of training data, other strategies like improved feature extraction method or specific data augmentation could be utilised in order to handle visually similar signs better.
- Pedestrian and Crossing Signs (Group 6) & Traffic Management Signs (Group 7):** On the other hand, the model was most accurate on the recognition of pedestrians and crossing signs achieving 95.83% accuracy and traffic management and directional signs at 95.56% accuracy. When it comes to the differences of the signs, they are, in general, more distinguishable in their visuals. That is probably, why the model felt more certain in defining these signs as the right ones. This high accuracy in these groups shows that the model is well suited for learning clear simple and well-defined patterns of the visual input.

- **Warning Signs (Group 5):** The worst performance was observed in the category of warning signs, which is a broad class that includes all symbols or signs in the form of patterns, and it scored 75.40% accuracy. This category might have increased complexity because the model had to learn a wide range of visual patterns. The performance was still acceptable, although this may mean that there is still potential to improve the way more flexible sign categories are handled.

6.4 Limitations

Despite the promising results achieved by the final model, several limitations were identified during the research that impacted its performance, particularly in terms of generalization and real-time prediction consistency.

1. Dataset

Imbalance:

Among the problems it was necessary to state that the most severe one was dataset imbalance. Some of the specific speed limit signs constituted a few examples of the total numbers while others like the pedestrian or the crossing signs were many. This affected the model's performance when it comes towards generalizing the identified traffic sign categories. Nevertheless, the lack of representation of such classes may have an impact on the ability of the data augmentation approach to teach the model to recognize these signs more frequently.

2. Generalization

Across

Regions:

Although the model achieved good accuracy for specific datasets, the test with the mixed dataset demonstrated the issue of varying regions generalization. The model trained on the German dataset was the best generalized, but the training of the model on a single dataset affected the performance badly when the signs of other regions were input to the model. As such, the future research should aim at developing a better integrated dataset comprising traffic signs from different areas as a way of establishing better generalization across the various environments.

3. Real-Time

Prediction

Inconsistencies:

This real-time experiment seemed to further show the model being unreliable

when it is applied to moving video streams. Other issues due to differences in lighting conditions, changes in camera position, and interference in surrounding sounds affected the model's reliability of its outcome predictions from one frame to another one. This limitation is actually indicating that the model needs to be stabilized and made reliable for real-world conditions by incorporating more techniques in the future which might include temporal smoothing and video-based contextual analysis.

4. **Computational Complexity and Resource Constraints:**

The training process of complex models especially when the approaches such as data augmentation and hyperparameter optimization are involved was computationally expensive. Due to hardware constraints, there was a limitation placed on how far they could take the system and the level of optimization which it was possible to attain especially when it came to real-time testing. These constraints may have also prevented the opportunity to spend more time in developing more complex models as well as to introduce more enhancements.

7. Future Work

The limitations identified in this study highlight several opportunities for future work to enhance the model's performance and applicability in real-world scenarios. Below are key areas that could be explored in subsequent research:

1. **Enhanced Real-Time Detection:**

Further research could potentially address the enhanced utilization of the model to work in real-time identification, especially in mobile environments. The addition of more advanced algorithms in tracking can make the model retrieve results more responsive and precise when dealing with dynamic environment examples like moving cars. This would improve the model's performance in detecting signs across situations and in low light conditions as seen in real life.

2. **Dataset Expansion:**

To enhance the model flexibility and performance, we should extend the database increasing the traffic signs number from other regions and zones. Increasing the

model's traffic sign recognition database with traffic signs of other countries and regions would also make the model to perform better in traffic systems of other regions, making it more generalized.

3. **Make** **Interactive** **UI:**

Designing a friendly graphical user interface (GUI) help to improve an environment where users feed images into the model for the traffic sign detection. This UI could help the users correct the model when it gives multiple wrong predictions so that the model weights can be balanced in real time situations. This would make the system more flexible and finer tune it overall with the help of the users.

4. **Model** **Ensemble:**

Applying Ensemble Learning and transfer learning might enhance the model's performance and increase the model's accuracy. Hence, there is a motivation to look for different models and apply ensemble techniques to produce a higher performing and more reliable traffic sign recognition system. Fine tuning of transfer learning could enhance the detection accuracy especially in classes of traffic signs which are rarity in developing countries.

5. **Continuous** **Hyperparameter** **Tuning:**

The optimization of the model includes the fine-tuning of hyperparameters. Further research could be done using more recent hyperparameter optimization techniques to enhance the performance of the model. There are several more adaptive hyperparameter tuning methods which could be incorporated in this model to ensure that the model's efficiency is maintained where there is new data and changing circumstances.

6. **Publish on the Cloud and Incorporate Continuous Training:**

Deploying the model on the cloud and integrating the idea of continuous training would make the model real and approximate the real world by training the model on new traffic sign over a period of time. Cloud deployment would allow the model to handle large volumes of data and adapt incrementally, as it was exposed to new signs and conditions. It would make the model more robust and adaptable when dealing with the real-world problems.

7. Transfer Learning on YOLOv6 with Better Image Quality:

Introducing transfer learning on the YOLOv6 model, known for its high-speed object detection capabilities, could further improve the detection rate, especially on augmented datasets. This would be especially useful for such cases where detection is time sensitive and needs to be as accurate as possible. Higher quality images could also be adopted in the training process as this would improve the model recognition of the signs in different settings.

8. Road and Sign Recognition in an Autonomous Miniature Car:

The model could be further developed to an independent miniature car, which could be seen as a rather promising and suitable area for further research. This project could entail integration of road detection in combination with the traffic sign recognition in order to achieve complete traffic and path recognition. It would be possible to test the model in a more controlled environment with an autonomous miniature car thus giving an indication of how well the complete traffic navigation system will function in a real application before proceeding to integrate it to larger Autonomous cars.

9. Multi-Task Learning:

For possible improvements of the project in the future, more advanced features such as Multi-Task Learning which can learn not only the traffic signs but also other features like vehicles, lanes, or traffic lights can be trained. In the case of autonomic driving, it is beneficial to put into one model several related tasks that could be solved in parallel making the system more efficient. It can help enhance the total system performance by integrating the representations of multiple tasks and thus, the workload of computation; meanwhile, such approach will eventually contribute to the construction of a more powerful and flexible driving assistance system.

8. Conclusion

The present study has proposed a Convolutional Neural Network (CNN) based Traffic Sign Recognition System which is able to handle the dataset collected from different regions like Germany, India and Asia. This project illustrated the use of CNNs for successfully recognizing traffic-sign information and at training time, no data set can be found comparable to German Traffic Sign Recognition Benchmark (GTSRB) on robustness nor generalization. It was further improved by performing data augmentation, balancing and hyperparameter tuning with ultimate test accuracy of 84.41%. Those adjustments ensured that the model was not only accurate, but overfitting had been corrected and it could generalize well beyond just fitting exactly to how it learned from training.

Conclusion One of the significant discoveries from this research is that dataset diversity and scale are very vital when training deep learning models for vehicle traffic sign identification. The German dataset — which was the most diverse and largest in regard to different examples, allowed for significantly better performance compared to the smaller and less varied Asian or Indian datasets. This being said using a big set of data that is as diverse (thus representing most real-life traffic signs and conditions) is needed. This combined dataset testing exercise also highlighted the drawback of training models on either regional dataset which results in poor generalization requiring for a comprehensive traffic sign dataset that encompasses signs from several regions.

This added new challenges in testing the accuracy of real-time predictions and showed some practical issues for deploying traffic sign recognition system on a live environment, which were not covered by previous methods. While the final model achieved remarkable results when tested against static images, it did not perform as well in live scenarios with elements like different light intensities and angles of capture were inconsistent due to environmental factors. These discrepancies highlight the need for more research to develop better performing models in terms of robustness and reliability when it comes to real-time traffic sign detection, especially as they would be crucial components in autonomous driving.

Additionally, this work highlighted the computational needs of training sophisticated CNN models. For instance, hyperparameter tuning using data augmentation and other techniques are computationally expensive operations for significant sized models which may limit experimentation as well optimization. Further research in this area could also target creating models that are more efficient (compromise on less accuracy) but significantly reduce the operational costs, making it well suited for deployment into resource-constrained devices as we see in case of autonomous vehicle.

However, certain challenges were met in the course of this research, and the following are the main contributions that this study has made to the field of traffic sign recognition. Firstly, the signs that are classified into different groups were observed and it was seen that when the signs are dissimilar and clearly distinct such as traffic signs with different shape, colour or phrasing, the model worked well on the other hand when the common features between them were high such as the speed limit sign images then the model didn't perform well. Such challenges call for improved feature extraction methods and specific data augmentation methods to be used in future application of such models.

Further, opportunities to enhance accurate and robust traffic sign recognition may be derived from exploring ensemble learning and transfer learning methodologies. These techniques can be used to develop more robust models capable of performing well across various scenarios and identifying a wider range of traffic signs with a significant degree of accuracy. Furthermore, subsequent dynamic interfaces of the user for operation and effective training procedures would enable the model to update its responses based on the real-life feedback received in the future.

As a result, the findings of this research open the door for potential future studies which can expand from the groundwork laid in this thesis. Improving real-time detection mechanisms, adding more data, and incorporation of the model in more elaborate self-driving systems are the next steps towards ensuring that traffic sign recognition models can easily be implemented in practice. Furthermore, applying it in practical tests with self-driving cars will allow for testing its performance in real-world scenarios and tuning it for real-world usage.

Therefore, this thesis gives a valuable input to the development of autonomous vehicle technologies and provides enlightening insights and future directions for CNN-based traffic sign recognition, their application to real-time scenarios, as well as their future improvements. With the advancement in the autonomous driving system, there is a need to develop reliable and flexible traffic sign recognition systems that will be extremely useful in manoeuvring through traffic systems in the various parts of the world.

References:

These references are in IEEE format.

Literature review:

A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Traffic Sign Recognition with Multi-Scale Convolutional Networks," in Proceedings of the 25th International Conference on Neural Information Processing Systems, 2012, pp. 1097-1105.

J. Stallkamp, M. Schlipsing, J. Salmen, and C. Igel, "Real-Time Traffic Sign Recognition from Video by Class-Specific Discriminative Features," in IEEE Transactions on Intelligent Transportation Systems, vol. 13, no. 4, pp. 1486-1498, Dec. 2012.

C. Chen, A. Seff, A. Kornhauser, and J. Xiao, "Traffic Sign Classification with Dataset Augmentation and Convolutional Neural Network," in Proceedings of the IEEE International Conference on Computer Vision, 2015, pp. 11-19.

S. Houben, J. Stallkamp, J. Salmen, M. Schlipsing, and C. Igel, "Robust Real-Time Traffic Sign Detection Using a Probabilistic Fusion Method," in IEEE Transactions on Intelligent Transportation Systems, vol. 15, no. 4, pp. 1570-1579, Aug. 2014.

C. Møgelmoose, M. M. Trivedi, and T. B. Moeslund, "Multiple-Dataset Traffic Sign Classification with OneCNN," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops, 2013, pp. 132-138.

T. Y. Tang, S. M. Tang, Y. Zhang, Y. Liu, and Z. Zhang, "Issues and Future Directions in Traffic Classification," in IEEE Communications Magazine, vol. 55, no. 7, pp. 37-43, Jul. 2017.

D. Ciregan, U. Meier, and J. Schmidhuber, "CNN Based Traffic Sign Classification Using Adam Optimizer," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 171-178.

M. Mathias, R. Timofte, R. Benenson, and L. Van Gool, "A Practical Approach for Detection and Classification of Traffic Signs Using Convolutional Neural Networks," in Proceedings of the IEEE Intelligent Vehicles Symposium, 2013, pp. 264-269.

S. Ioffe and C. Szegedy, "Traffic-Sign Detection and Classification in the Wild," in Proceedings of the IEEE International Conference on Computer Vision Workshops, 2016, pp. 51-59.

P. Sermanet and Y. LeCun, "Traffic Sign Classification Using Deep Neural Network," in Proceedings of the IEEE International Joint Conference on Neural Networks, 2011, pp. 281-288.

Additional Reference papers & conference:

A. Faisal, M. Kamruzzaman, T. Yigitcanlar, and G. Currie, "Understanding Autonomous Vehicles," Journal of Transport and Land Use, vol. 12, no. 1, pp. 45-72, 2019.

M. Martínez-Díaz and F. Soriguera, "Autonomous Vehicles: Theoretical and Practical Challenges," in Transportation Research Procedia, vol. 33, pp. 275-282, 2018.

N. Youssouf, "Traffic Sign Classification Using CNN and Detection Using Faster-RCNN and YOLOV4," Heliyon, vol. 8, no. 12, 2022.

S. Liu, "A Traffic Sign Image Recognition and Classification Approach Based on Convolutional Neural Network," in 2019 11th International Conference on Measuring Technology and Mechatronics Automation (ICMTMA), IEEE, 2019, pp. 43-47.

C. G. Serna and Y. Ruichek, "Classification of Traffic Signs: The European Dataset," IEEE Access, vol. 6, pp. 78136-78148, 2018.